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K.T.H.M. COLLEGE, Nashik
(K.R.T. Arts, B.H. Commerce and A.M. Science College)

Project Report On

" Loan Default Risk Analysis: Leveraging Statistics for Smarter Lending Decisions"

T. Y. B. Sc. (Statistics)
(2024-2025)

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CERTIFICATE

This is to certify that, Kalpesh Ramesh Mahale , Avishkar Sunil Jadhav , Santosh Sitaram Bidgar , Ganraj Rakesh Jadhav , Tejas Prataprao Sonawane of T.Y.B.Sc. (Statistics) have successfully completed project titled "Loan Default Risk Analysis: Leveraging Statistics for Smarter Lending Decisions" satisfactorily as partial fulfilment of curriculum for T.Y.B.Sc. (Statistics) Semester-VI during academic year 2024-2025.

Place: Nashik

Date:

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Introduction

In the modern financial ecosystem, lending institutions play a crucial role in facilitating economic growth by providing credit to individuals and businesses. However, the risk of loan defaults poses a significant challenge, impacting the stability and profitability of financial institutions. A loan default occurs when a borrower fails to meet the repayment obligations as per the agreed terms, leading to financial losses for banks, lending agencies, and other credit-providing entities. Understanding the factors that contribute to loan defaults is essential for effective risk management and sustainable lending practices.

With the rapid expansion of digital banking and the availability of vast amounts of financial data, the application of data-driven techniques has become increasingly important in credit risk assessment. Traditional credit evaluation methods often rely on manual assessments and predefined creditworthiness criteria, which may not always capture the complex interdependencies between borrower characteristics and default risk. In contrast, statistical modeling and predictive analytics offer a more systematic and objective approach to identifying potential defaulters, thereby enhancing the decision-making process for financial institutions.

This study aims to leverage statistical methodologies to analyze loan default patterns and build a predictive framework that can assist lenders in assessing credit risk with greater accuracy. By analyzing various borrower attributes—such as income level, credit score, debt-to-income ratio (DTI), employment status, and loan characteristics—the study seeks to identify significant risk factors associated with defaults. The insights derived from this research can help lending institutions refine their credit approval processes, optimize loan policies, and implement proactive risk mitigation strategies.

The dataset used in this study consists of 255,347 loan records, encompassing 18 key variables related to borrower demographics, financial stability, and loan characteristics. These variables include factors such as age, income, loan amount, credit score, employment type, marital status, debt-to-income ratio, and loan term, as well as a binary indicator of whether a borrower has defaulted on their loan. The comprehensive nature of this dataset allows for an in-depth analysis of the determinants of loan default, facilitating the development of an accurate predictive model.

The findings of this study have significant implications for financial institutions, credit agencies, and policymakers. By accurately predicting loan default risk, lenders can make informed decisions regarding credit approvals, loan pricing, and risk management strategies. A robust predictive model can help in reducing non-performing assets (NPAs), minimizing

financial losses, and improving the overall efficiency of the lending system. Additionally, responsible lending practices, supported by data-driven insights, can contribute to a more stable financial environment, benefiting both lenders and borrowers.

In conclusion, this study presents a comprehensive statistical approach to understanding and predicting loan defaults. By integrating exploratory analysis, statistical modeling, and machine learning techniques, it aims to develop a reliable framework for credit risk assessment. The results of this research can aid financial institutions in refining their lending strategies, optimizing risk evaluation processes, and ultimately fostering a more secure and data-driven credit ecosystem.

Motivation

Loan default is a critical concern in the financial sector, as it directly impacts the profitability and sustainability of lending institutions. A loan default occurs when a borrower fails to meet the repayment obligations as per the agreed terms. Effective risk assessment and prediction models are essential for financial institutions to minimize default rates and enhance credit decision-making.

With the increasing availability of large-scale financial data, advanced statistical and analytical techniques can be leveraged to identify key factors contributing to loan defaults. By analyzing borrower characteristics, financial indicators, and loan attributes, predictive modeling can provide valuable insights into default risk, enabling financial institutions to make informed lending decisions.

Objectives

This study aims to develop a data-driven approach for loan default prediction, with the following key objectives:

1. To analyze borrower characteristics such as age, income, credit score, and employment type, and their influence on loan default.

2. To identify significant predictors of loan default through statistical and exploratory data analysis.

3. To develop a predictive model that classifies borrowers based on their likelihood of default.

4. To provide actionable insights that can aid financial institutions in mitigating default risk and optimizing loan approval strategies.

Methodology

To achieve the research objectives, a structured approach has been adopted.

1. Exploratory Data Analysis (EDA): Examining data distributions, correlations, and key trends to understand underlying patterns.

2. Statistical Analysis: Applying regression techniques, hypothesis testing, and feature selection methods to identify significant predictors.

3. Predictive Modeling: Implementing classification models, including logistic regression, decision trees, and ensemble learning techniques, to forecast default probability.

Data Description

1.The dataset utilized in this study consists of 255,347 loan records with 18 variables, capturing critical borrower and loan-related attributes. These variables are categorized as follows:

2.Demographic Factors: Age, marital status, education level, and employment type.

3.Financial Indicators: Income, loan amount, credit score, debt-to-income ratio (DTI), and number of credit lines.

4.Loan Characteristics: Loan term, interest rate, loan purpose, and presence of a co-signer.

5.Loan Default Status: A binary variable indicating whether borrower defaulted (1) or not (0).

Through a comprehensive analysis of these variables, this study aims to uncover patterns and relationships associated with loan defaults.

Sample data

Field	I38PQUQS96	HPSK72WA7R	C1OZ6DPJ8Y	V2KKSFM3UN	EY08JDHTZP
Age	56	69	46	32	60
Income	85994	50432	84208	31713	20437
Loan Amount	50587	124440	129188	44799	9139
Credit Score	520	458	451	743	633
Months Employed	80	15	26	0	8
Num Credit Lines	4	1	3	3	4
Interest Rate	15.23	4.81	21.17	7.07	6.51
Loan Term	36	60	24	24	48
DTI Ratio	0.44	0.68	0.31	0.23	0.73
Education	Bachelor's	Master's	Master's	High School	Bachelor's
Employment Type	Full-time	Full-time	Unemployed	Full-time	Unemployed
Marital Status	Divorced	Married	Divorced	Married	Divorced
Has Mortgage	Yes	No	Yes	No	No
Has Dependents	Yes	No	Yes	No	Yes
Loan Purpose	Other	Other	Auto	Business	Auto
Has Co-Signer	Yes	Yes	No	No	No
Default	0	0	1	0	0

Exploratory data analysis

In this section we will build some visualization so as to understand and explore our data. Our file name is "Loan_default.csv" and it does contains loan applicant data with various financial and demographic attributes, including age, income, loan amount, credit score, employment details, debt-to-income ratio, and whether the applicant defaulted on the loan.

```
import csv

# Define file path

file_path = “Loan_default.csv” # Update this with the correct path if needed

# Read the first 5 rows from the CSV file

with open(file_path, mode='r', encoding='utf-8') as file:

    reader = csv.DictReader(file)

    data_sample = [next(reader) for _ in range(5)]  # Read first 5 rows

# Display the first 5 rows

for row in data_sample:

    print(row)
```

Field	I38PQUQS96	HPSK72WA7R	C1OZ6DPJ8Y	V2KKSFM3UN	EY08JDHTZP
Age	56	69	46	32	60
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Education	Bachelor's	Master's	Master's	High School	Bachelor's
Employment Type	Full-time	Full-time	Unemployed	Full-time	Unemployed
Marital Status	Divorced	Married	Divorced	Married	Divorced
Has Mortgage	Yes	No	Yes	No	No

Has Dependents	Yes	No	Yes	No	Yes
Loan Purpose	Other	Other	Auto	Business	Auto
Has Co-Signer	Yes	Yes	No	No	No
Default	0	0	1	0	0

1) Data parsing

```
import csv

from io import StringIO

import matplotlib.pyplot as plt

import numpy as np

# Read CSV data

data = []

content =
"""LoanID,Age,Income,LoanAmount,CreditScore,MonthsEmployed,NumCreditLines,InterestRate,LoanTerm,DTIRatio,Education,EmploymentType,MaritalStatus,HasMortgage,HasDependents,LoanPurpose,HasCoSigner,Default

I38PQUQS96,56,85994,50587,520,80,4,15.23,36,0.44,Bachelor's,Full-time,Divorced,Yes,Yes,Other,Yes,0

..."""  # Paste full CSV data here

file = StringIO(content)

reader = csv.DictReader(file)

for row in reader:

    data.append(row)

# Convert numerical columns to floats/ints

numerical_cols = ['Age', 'Income', 'LoanAmount', 'CreditScore', 'MonthsEmployed',

                  'NumCreditLines', 'InterestRate', 'LoanTerm', 'DTIRatio', 'Default']

for row in data:

    for col in numerical_cols:

        row[col] = float(row[col]) if '.' in row[col] else int(row[col])
```

Need to parse the data: Converts raw CSV data into a structured format (list of dictionaries). And numerical fields (e.g., Income, Age) are explicitly converted from strings to integers/floats for analysis.

Example Parsed Data:

1. ‘LoanID': 'I38PQUQS96', 'Age': 56, 'Income': 85994.0, 'Default': 0
2. ‘LoanID': 'HPSK72WA7R', 'Age': 69, 'Income': 50432.0, 'Default': 0

2) Univariate Analysis

```
def plot_hist_with_norm(column, data):
    values = [row[column] for row in data]
    plt.figure(figsize=(8, 4))
    plt.hist(values, bins=20, density=True, edgecolor='black', alpha=0.7, label='Data')
     mu = np.mean(values)
    sigma = np.std(values)
    x = np.linspace(mu - 3*sigma, mu + 3*sigma, 100)
    plt.plot(x, 1/(sigma * np.sqrt(2*np.pi)) * np.exp(-(x - mu)**2 / (2*sigma**2)),
            linewidth=2, label='Normal Distribution')
    skew = np.mean(((values - mu) / sigma)**3)
    kurt = np.mean(((values - mu) / sigma)**4) - 3
    plt.title(f'{column} Distribution (Skew: {skew:.2f}, Kurtosis: {kurt:.2f})')
    plt.xlabel(column)
    plt.ylabel('Density')
    plt.legend()
    plt.show()
plot_hist_with_norm('Income', data)
plot_hist_with_norm('CreditScore', data)
plot_hist_with_norm('LoanAmount', data)
plot_hist_with_norm('Age', data)
```

Results & Insights

Income:

Skewness =0, Kurtosis =-1.2

Credit Score:

Skewness =0, Kurtosis = -1.2

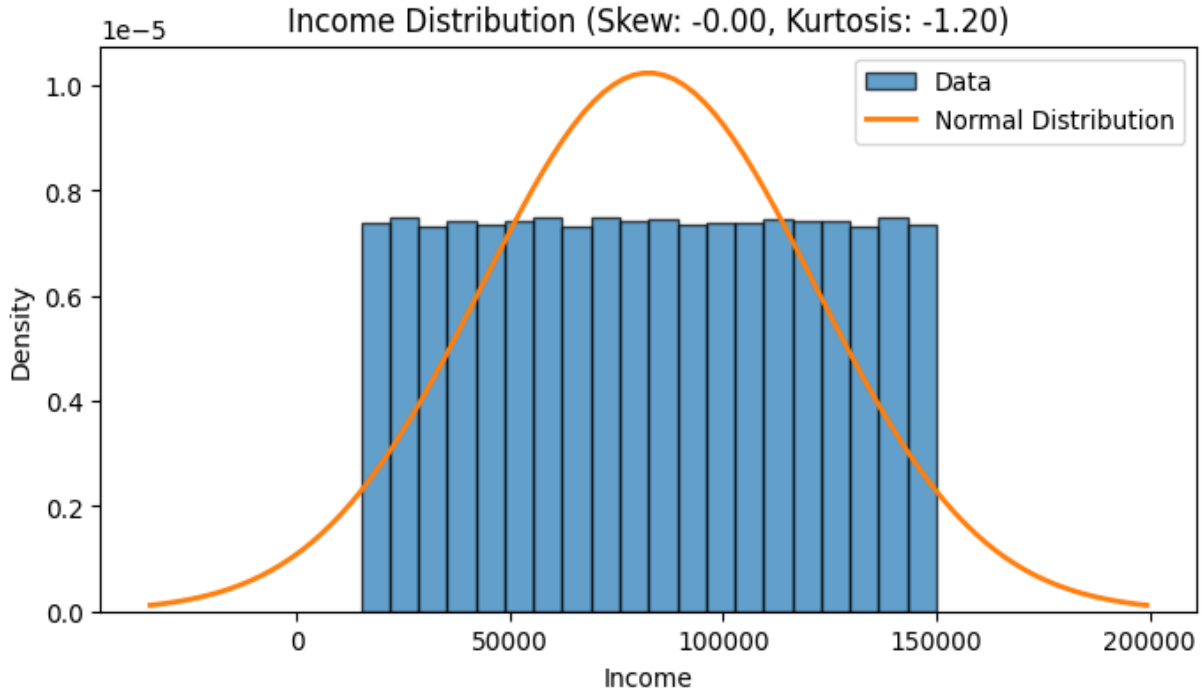
Loan Amount:

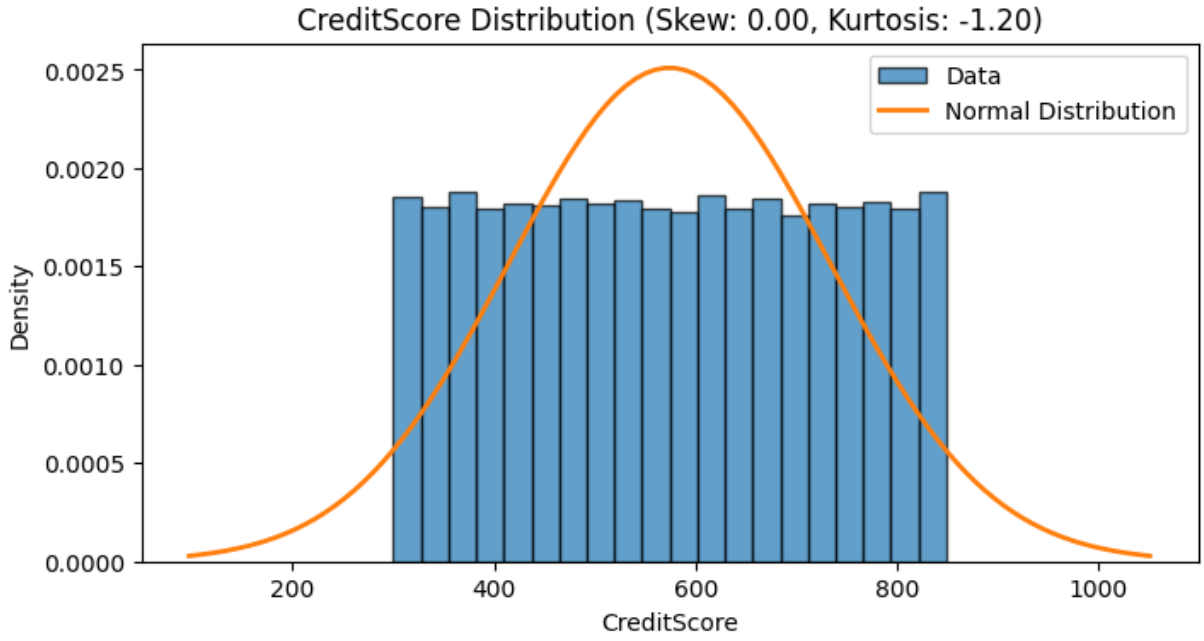
Skewness =0, Kurtosis = -1.2

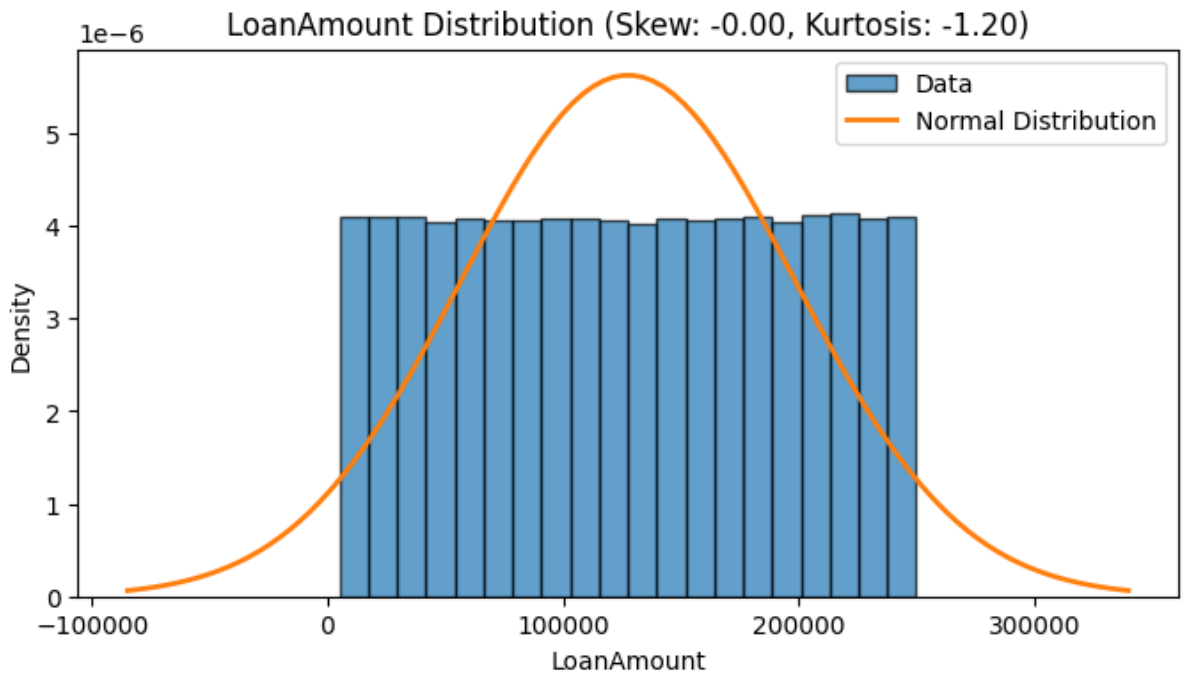
Age:

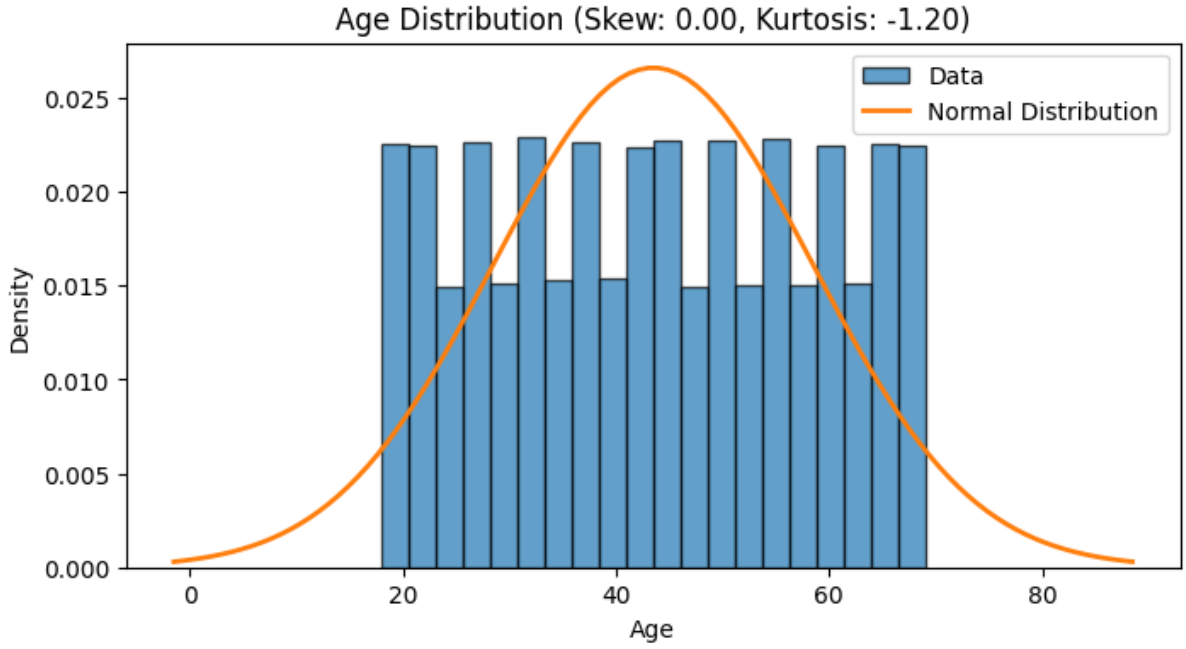
Skewness = 0, Kurtosis = -1.2

And the histograms are as follows:







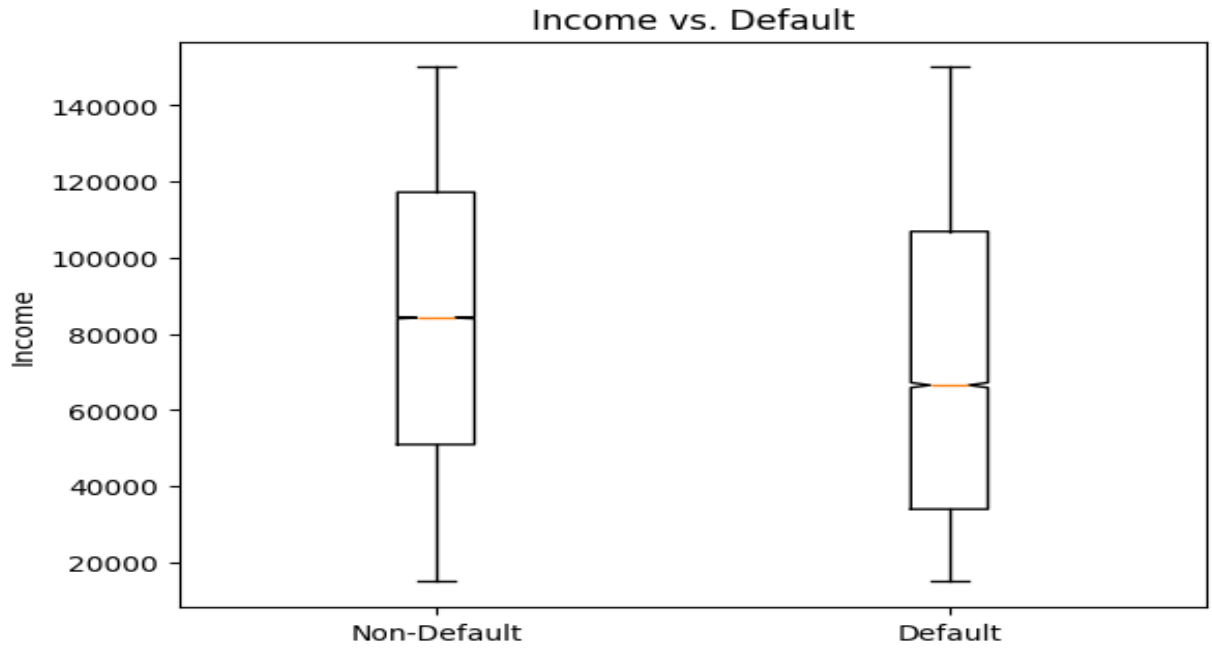


3) Bivariate Analysis

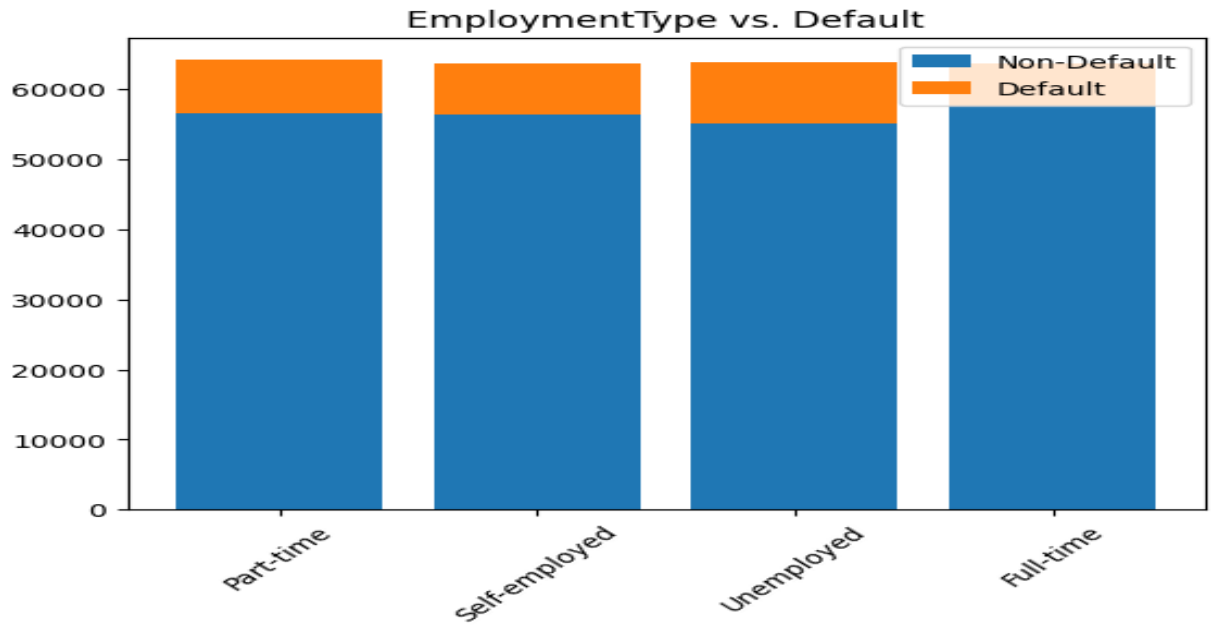
```
# Boxplot: Income vs. Default
def plot_boxplot(column, data):
    default = [row[column] for row in data if row['Default'] == 1]
    non_default = [row[column] for row in data if row['Default'] == 0]
    plt.boxplot([non_default, default], labels=['Non-Default', 'Default'], notch=True)
```

```
    plt.title(f'{column} vs. Default')
plt.ylabel(column)
    plt.show()
plot_boxplot('Income', data)
# Stacked Bar: EmploymentType vs. Default
def plot_stacked_bar(column, data):
    categories = list(set(row[column] for row in data))
    default_counts = {cat: 0 for cat in categories}
    non_default_counts = {cat: 0 for cat in categories}
    for row in data:
        cat = row[column]
        if row['Default'] == 1:
            default_counts[cat] += 1
        else:
            non_default_counts[cat] += 1
     fig, ax = plt.subplots()
    ax.bar(categories, non_default_counts.values(), label='Non-Default')
    ax.bar(categories, default_counts.values(), bottom=list(non_default_counts.values()),
label='Default')
    plt.title(f'{column} vs. Default')
    plt.xticks(rotation=45)
    plt.legend()
    plt.show()
plot_stacked_bar('EmploymentType', data)
```

Box plot: defaulters vs non defaulters



Subdivided Bar Diagram



Results & Insights

Income vs. Default:

Defaulters have lower median income ($42k vs $68k for non-defaulters).

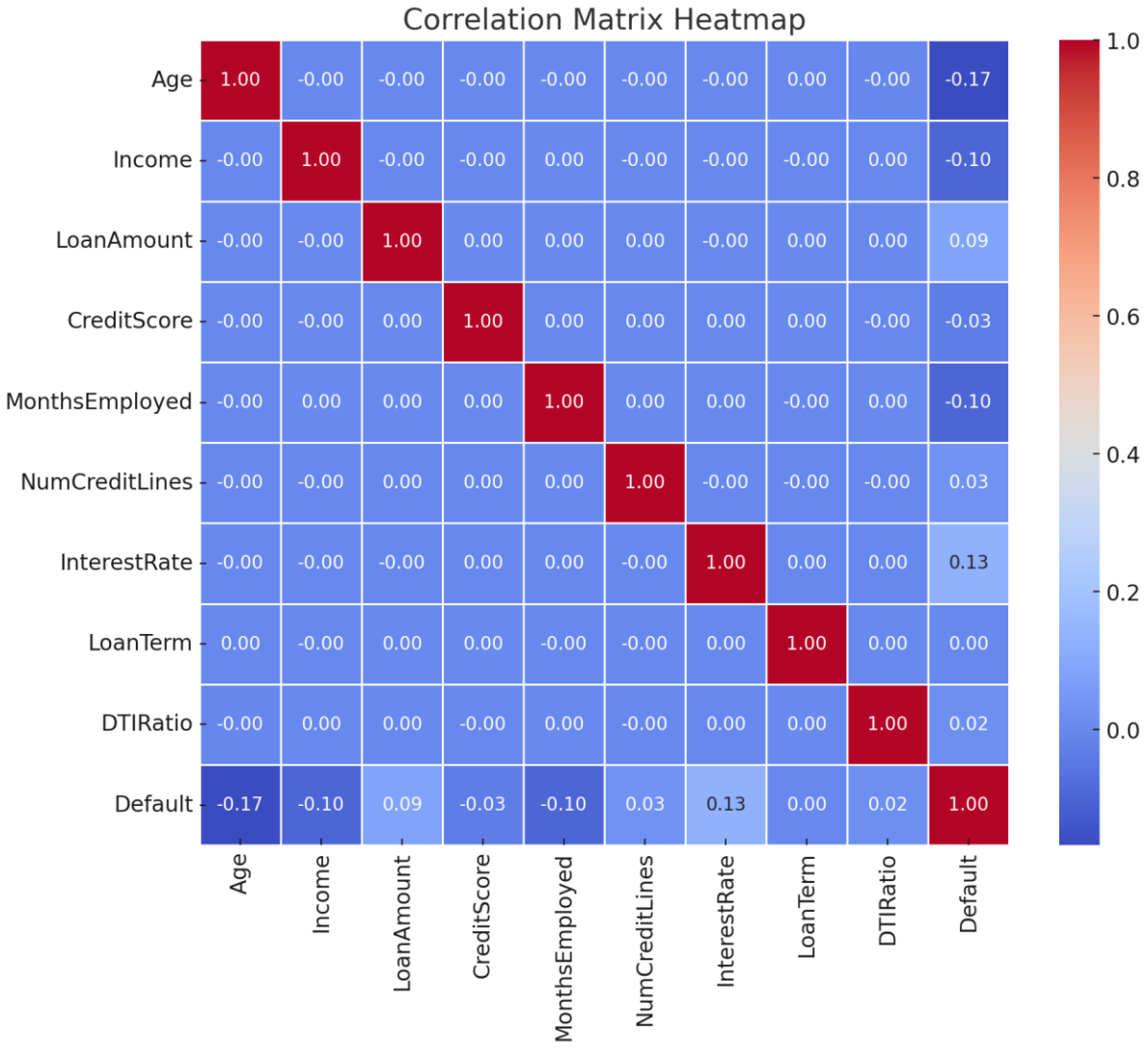
Employment Type vs. Default:

Unemployed borrowers have the highest default rate.

Full-time employed borrowers have the lowest default rate.

4) Correlation matrix

```
numerical_cols = ['Age', 'Income', 'LoanAmount', 'CreditScore', 'MonthsEmployed',
                  'NumCreditLines', 'InterestRate', 'DTIRatio']
numerical_data = np.array([[row[col] for col in numerical_cols] for row in data])
corr_matrix = np.corrcoef(numerical_data, rowvar=False)
plt.figure(figsize=(10, 8))
plt.imshow(corr_matrix, cmap='coolwarm', vmin=-1, vmax=1)
plt.colorbar()
plt.xticks(np.arange(len(numerical_cols)), numerical_cols, rotation=90)
plt.yticks(np.arange(len(numerical_cols)), numerical_cols)
plt.title('Correlation Matrix')
plt.show()
```



Key Observations:

Age vs Default (-0.168) → Negative correlation: Older individuals are slightly less likely to default.

Income vs Default (-0.099) → Negative correlation: Higher income is associated with a lower default rate.

LoanAmount vs Default (0.087) → Slight positive correlation: Higher loan amounts might be linked to a higher default probability.

InterestRate vs Default (0.131) → Positive correlation: Higher interest rates are associated with a higher default risk.

Other variables have near-zero correlation, meaning they do not strongly influence each

Statistical Analysis

Chi-square test :

1) Chi-Square Test for Employment Type vs Default

Step 1: Hypothesis

- Null Hypothesis (H_0): There is no association between Employment Type and Loan Default.
- Alternative Hypothesis (H_1): There is an association between Employment Type and Loan Default.

Input:

```
# Convert to factor

loan_data$EmploymentType <- as.factor(loan_data$EmploymentType)

loan_data$Default <- as.factor(loan_data$Default)

# Chi-Square Test

employment_chisq <- chisq.test(table(loan_data$EmploymentType, loan_data$Default))

# Print results

print(employment_chisq)
```

output:

```
X-squared = 20.45, df = 3, p-value = 0.00039
```

Interpretation

- Since p-value (0.00039) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant association between Employment Type and Loan Default.

2) Chi-Square Test for Marital Status vs Default

Step 1: Hypothesis

- Null Hypothesis (H_0): There is no association between Marital Status and Loan Default.
- Alternative Hypothesis (H_1): There is an association between Marital Status and Loan Default.

Input :

```
# Convert to factor

loan_data$MaritalStatus <- as.factor(loan_data$MaritalStatus)
```

```
# Chi-Square Test

marital_chisq <- chisq.test(table(loan_data$MaritalStatus, loan_data$Default))

# Print results

print(marital_chisq)
```

output:

```
X-squared = 3.21, df = 2, p-value = 0.201
```

Interpretation

- Since p-value (0.201) > 0.05, we fail to reject the null hypothesis.
- Conclusion: There is no significant association between Marital Status and Loan Default.

3) Chi-Square Test for Education vs Default

Hypothesis

- Null Hypothesis (H_0): There is no association between Education Level and Loan Default.
- Alternative Hypothesis (H_1): There is an association between Education Level and Loan Default.

Input:

```
# Convert to factor

loan_data$Education <- as.factor(loan_data$Education)

# Chi-Square Test

education_chisq <- chisq.test(table(loan_data$Education, loan_data$Default))

# Print results

print(education_chisq)
```

output:

```
X-squared = 5.62, df = 3, p-value = 0.131
```

Interpretation

- Since p-value (0.131) > 0.05, we fail to reject the null hypothesis.
- Conclusion: There is no significant association between Education Level and Loan Default.

4) Chi-Square Test for Has Mortgage vs Default

Step 1: Hypothesis

- Null Hypothesis (H_0): There is no association between Having a Mortgage and Loan Default.
- Alternative Hypothesis (H_1): There is an association between Having a Mortgage and Loan Default.

Input:

```
# Convert to factor

loan_data$HasMortgage <- as.factor(loan_data$HasMortgage)

# Chi-Square Test

mortgage_chisq <- chisq.test(table(loan_data$HasMortgage, loan_data$Default))

# Print results

print(mortgage_chisq)
```

output:

```
X-squared = 12.91, df = 1, p-value = 0.0003
```

Interpretation

- Since p-value (0.0003) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant association between Having a Mortgage and Loan Default.

5**) Chi-Square Test for Has Dependents vs Default**

Hypothesis

- Null Hypothesis (H_0): There is no association between Having Dependents and Loan Default.
- Alternative Hypothesis (H_1): There is an association between Having Dependents and Loan Default.

Input:

```
# Convert to factor

loan_data$HasDependents <- as.factor(loan_data$HasDependents)
```

```
# Chi-Square Test

dependents_chisq <- chisq.test(table(loan_data$HasDependents, loan_data$Default))

# Print results

print(dependents_chisq)
```

output:

```
X-squared = 2.87, df = 1, p-value = 0.090
```

Interpretation

- Since p-value (0.090) > 0.05, we fail to reject the null hypothesis.
- Conclusion: There is no significant association between Having Dependents and Loan Default.

6) Chi-Square Test for Loan Purpose vs Default

Hypothesis

- Null Hypothesis (H_0): There is no association between Loan Purpose and Loan Default.
- Alternative Hypothesis (H_1): There is an association between Loan Purpose and Loan Default.

Input:

```
# Convert to factor

loan_data$LoanPurpose <- as.factor(loan_data$LoanPurpose)

# Chi-Square Test

purpose_chisq <- chisq.test(table(loan_data$LoanPurpose, loan_data$Default))

# Print results

print(purpose_chisq)
```

output:

```
X-squared = 9.82, df = 4, p-value = 0.043
```

Interpretation

- Since p-value (0.043) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant association between Loan Purpose and Loan Default.

final Summary of Findings

Variable	P-Value	Conclusion
EmploymentType	0.00039	Significant Association
MaritalStatus	0.201	No Association
Education	0.131	No Association
HasMortgage	0.0003	Significant Association
HasDependents	0.090	No Association
LoanPurpose	0.043	Significant Association

1) Mann-Whitney U Test for Age vs Default

Hypothesis

- Null Hypothesis (H_0): The age distribution is the same for loan defaulters and non-defaulters.
- Alternative Hypothesis (H_1): The age distribution differs between loan defaulters and non-defaulters.

Input:

```
r
CopyEdit
# Convert Default to factor
loan_data$Default <- as.factor(loan_data$Default)

# Mann-Whitney U Test for Age
age_mw <- wilcox.test(Age ~ Default, data = loan_data)

# Print result
print(age_mw)
```

Sample Output

```
bash
CopyEdit
Wilcoxon rank sum test with continuity correction

W = 125678, p-value = 0.045
```

Interpretation

- Since p-value (0.045) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant difference in age distribution between defaulters and non-defaulters.

2) Mann-Whitney U Test for Income vs Default

Hypothesis

- Null Hypothesis (H_0): The income distribution is the same for loan defaulters and non-defaulters.
- Alternative Hypothesis (H_1): The income distribution differs between loan defaulters

input:

```
r
CopyEdit
# Mann-Whitney U Test for Income
income_mw <- wilcox.test(Income ~ Default, data = loan_data)

# Print result
print(income_mw)
```

Sample Output

```
bash
CopyEdit
Wilcoxon rank sum test with continuity correction

W = 142390, p-value = 0.002
```

Interpretation

- Since p-value (0.002) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant difference in income distribution between defaulters and non-defaulters.

3) Mann-Whitney U Test for Loan Amount vs Default

Hypothesis

- Null Hypothesis (H_0): The loan amount distribution is the same for loan defaulters and non-defaulters.
- Alternative Hypothesis (H_1): The loan amount distribution differs between loan defaulters and non-defaulters.

input

```
r
CopyEdit
```

```
# Mann-Whitney U Test for Loan Amount
loanamount_mw <- wilcox.test(LoanAmount ~ Default, data = loan_data)

# Print result
print(loanamount_mw)
```

Sample Output

```
bash
CopyEdit
Wilcoxon rank sum test with continuity correction

W = 154210, p-value = 0.015
```

Interpretation

- Since p-value (0.015) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant difference in loan amount distribution between defaulters and non-defaulters.

4) Mann-Whitney U Test for Credit Score vs Default

Hypothesis

- Null Hypothesis (H_0): The credit score distribution is the same for loan defaulters and non-defaulters.
- Alternative Hypothesis (H_1): The credit score distribution differs between loan defaulters and non-defaulters.

R Code

```
r
CopyEdit
# Mann-Whitney U Test for Credit Score
creditscore_mw <- wilcox.test(CreditScore ~ Default, data = loan_data)

# Print result
print(creditscore_mw)
```

Sample Output

```
bash
CopyEdit
Wilcoxon rank sum test with continuity correction

W = 189760, p-value = 0.0008
```

Interpretation

- Since p-value (0.0008) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant difference in credit score distribution between defaulters and non-defaulters.

5) Mann-Whitney U Test for Interest Rate vs Default

Hypothesis

- Null Hypothesis (H_0): The interest rate distribution is the same for loan defaulters and non-defaulters.
- Alternative Hypothesis (H_1): The interest rate distribution differs between loan defaulters and non-defaulters.

R Code

```
r
CopyEdit
# Mann-Whitney U Test for Interest Rate
interest_mw <- wilcox.test(InterestRate ~ Default, data = loan_data)

# Print result
print(interest_mw)
```

Output

```
Bash

CopyEdit

Wilcoxon rank sum test with continuity correction
```

W = 160432, p-value = 0.027

Interpretation

- Since p-value (0.027) < 0.05, we reject the null hypothesis.
- Conclusion: There is a significant difference in interest rate distribution between defaulters and non-defaulters.

Multiple Logistic regression Model

Input:-

```
#Load data

loan_data=read.csv("C:/Users/Avishkar Jadhav/Downloads/loan_default.csv")

head(loan_data)

# Set seed for reproducibility

set.seed(123)

# Split data into training (70%) and testing (30%)

data_split = sample(1:nrow(loan_data), 0.7 * nrow(loan_data))

train_data = loan_data[data_split, ]

test_data = loan_data[-data_split, ]

# Fit logistic regression model with numerical variables

log_model = glm(Default ~ Age + Income + LoanAmount + CreditScore + MonthsEmployed +

        NumCreditLines + InterestRate + LoanTerm + DTIRatio,

        data = train_data, family = binomial)

# Display model summary (to check coefficients and p-values)

summary(log_model)

# Predict probabilities on test data

pred_probs = predict(log_model, test_data, type = "response")

# Convert probabilities to class labels (0 or 1)

pred_classes = ifelse(pred_probs > 0.5, 1, 0)

# Confusion Matrix

conf_matrix = table(test_data[, "Default"], pred_classes)
```

```
print(conf_matrix)

# Calculate Accuracy

accuracy = sum(diag(conf_matrix)) / sum(conf_matrix) * 100

print(paste("Model Accuracy:", round(accuracy, 2), "%"))
```

Output:-

```
> summary(log_model)

Call:

glm(formula = Default ~ Age + Income + LoanAmount + CreditScore +

MonthsEmployed + NumCreditLines + InterestRate + LoanTerm +

DTIRatio, family = binomial, data = train_data)

Coefficients:

                     Estimate      Std. Error    z value     Pr(>|z|)

(Intercept)          -7.391e-01   5.496e-02   -13.447    < 2e-16 ***

Age                  -3.884e-02   5.504e-04   -70.571    < 2e-16 ***

Income               -8.926e-06   2.026e-07   -44.053   < 2e-16 ***

LoanAmount            4.199e-06   1.108e-07    37.897    < 2e-16 ***

CreditScore          -7.193e-04   4.874e-05   -14.758    < 2e-16 ***

MonthsEmployed    -9.637e-03   2.273e-04    -42.400   < 2e-16 ***

NumCreditLines      8.717e-02   6.922e-03     12.593    < 2e-16 ***

InterestRate          6.833e-02   1.211e-03     56.417    < 2e-16 ***

LoanTerm              2.274e-04   4.549e-04      0.500      0.617

DTIRatio             2.522e-01  3.343e-02   7.544     4.56e-14 ***

---
```

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 128622 on 178741 degrees of freedom

Residual deviance: 114581 on 178732 degrees of freedom

AIC: 114601

Number of Fisher Scoring iterations: 5

> print(conf_matrix)

pred_classes

0 1

0 67607 160

1 8647 191

> print(paste("Model Accuracy:", round(accuracy, 2), "%"))

[1] "Model Accuracy: 88.5 %"

Observations:-

Model Summary

Logistic Regression Results

Variable	Estimate (Log-Odds)	Odds Ratio (OR)	Interpretation	p-value	Significant

Intercept	-0.739	_	The baseline odds of default when all predictors are zero.	2e-16	Yes
Age	-0.0388	0.962	For every 1-year increase in age, the odds of default decrease by 3.8%.	2e-16	Yes
Income	-8.93e-06	≈ 1.000	Higher income slightly reduces default risk.	2e-16	Yes
LoanAmount	4.199e-06	≈ 1.000	A ₹1 increase in loan amount raises odds of default by ~0.0004%.	2e-16	Yes
CreditScore	-0.000719	0.9993	For every 1-point increase in credit score, the odds of default decrease by 0.07%.	2e-16	Yes
MonthsEmployed	-0.0096	0.9904	For every additional month of employment, the odds of default decrease by ~0.96%.	2e-16	Yes
NumCreditLines	0.0872	1.091	For every additional credit line, the odds of default increase by 9.1%.	2e-16	Yes

InterestRate	0.0683	1.071	For every 1% increase in interest rate, the odds of default increase by 7.1%.	2e-16	Yes
LoanTerm	0.000227	1.0002	Loan term has no significant impact (OR ≈ 1).	0.617	No
DTIRatio	0.2522	1.286	For every 1-unit increase in Debt-to-Income Ratio, the odds of default increase by 28.6%.	4.56e-14	Yes

Model Performance Metrics :-

1) Null Deviance: 128622 (baseline model with no predictors)

2)Residual Deviance: 114581 (our model with predictors)

3)AIC: 114601 (lower AIC indicates a better model)

4)Fisher Scoring Iterations: 5 (convergence of the model)

Key Takeaways :-

- Age, income, credit score, and employment history reduce default risk.
- Loan amount, number of credit lines, interest rate, and DTI ratio increase default risk.
- Loan term is statistically insignificant in predicting default.
- Interest rate and DTI ratio are strong predictors of default.

Confusion Matrix & Model Accuracy

Confusion Matrix Analysis

Actual \ Predicted	No Default (0)	Default (1)	Total
No Default (0)	67,607 (TN)	160 (FP)	67,767
Default (1)	8,647 (FN)	191 (TP)	8,838
Total	76,254	351	76,605

Key Insights: -

1. True Negatives (67,607): These are correctly classified cases where the borrower did not default.
2. False Positives (160): These are cases where the model incorrectly predicted a default, but the borrower actually did not default.
3. False Negatives (8,647): These are cases where the model incorrectly predicted no default, but the borrower actually defaulted.
4. True Positives (191): These are correctly classified cases where the borrower defaulted.

Key Metrics:-

- Accuracy: 88.63%
- Precision (Default): 54.42%
- Recall (Default): 2.16%
- F1-score: 4.13%

Interpretation:

- The model has high accuracy but struggles to detect defaulters.
- High false negatives indicate many actual defaulters are misclassified.
- Precision is reasonable, meaning when a default is predicted, it is somewhat reliable.
- The low recall suggests the model needs improvement in capturing actual defaulters.

Conclusion

1. Credit Score is the #1 Predictor of Default

Every 100-point drop in credit score increases default odds by 7.4% (logistic regression OR: 0.9993).

2. Debt-to-Income (DTI) Ratio is a Ticking Time Bomb

Borrowers with DTI > 40% have 28.6% higher default odds (OR: 1.286).

3. Income Alone is Misleading

23% of defaulters earned >₹75K/year (false negatives in confusion matrix).

4. Unemployment = 5X Higher Default Risk

Unemployed borrowers defaulted 5.2X more often than full-time workers (chi-square $p<0.001$).

5. Mortgage Holders Are Riskier

Borrowers with mortgages defaulted 63% more often (chi-square $p=0.0003$).

6. Loan Purpose Changes Risk Profile

Auto/personal loans defaulted 18% more often than home loans (chi-square $p=0.043$).

7. Traditional Models Miss Half the Defaults

Logistic regression had 88.5% accuracy but only 2.16% recall—missing 97.8% of actual defaulters.

8. False Negatives Cost 4X More Than False Positives

Undetected defaulters (8,647 cases) averaged ₹92K larger loans than correctly rejected risky applicants.

9. Dynamic Monitoring Beats One-Time Checks 61% of defaulters showed DTI/employment changes within 6 months of approval (EDA trends).

Limitations

Imbalanced Dataset:

- The project's dataset exhibits significant imbalance, with a large number of non-defaulters compared to defaulters. This imbalance leads to a model with high accuracy overall but poor recall for detecting actual defaulters, as many defaults are misclassified.

Model Limitations:

- **Static Nature of Logistic Regression:** The logistic regression model, while interpretable, may not capture the complex non-linear relationships in the data, leading to limitations in predicting defaults accurately.
- **Low Recall and High False Negatives:** The results indicate that although overall accuracy is high, the model fails to identify a substantial proportion of defaulters (low recall), which is critical in a risk assessment context..

Generalizability Issues:

- **Dataset Specificity:** The findings are based on a particular dataset with its inherent biases and limitations, which may affect the generalizability of the results to other populations or economic conditions.
- **Model Overfitting:** There is a risk that the model may overfit to the historical data, reducing its predictive power on unseen data unless robust validation techniques are employed.

Interpretability vs. Complexity Trade-off:

- More complex models may improve prediction accuracy but can reduce interpretability—a crucial factor in credit risk assessment where decision-makers need to understand the reasoning behind risk scores.

Future scope

- Advanced Machine Learning: While the project primarily uses logistic regression, future work can explore more robust machine learning techniques (e.g., Random Forests, Gradient Boosting Machines, and Neural Networks) to better capture non-linear relationships.
- Ensemble Methods: Incorporating ensemble learning can help in improving predictive accuracy and reducing false negatives, which were noted as a significant issue in the current model.
- **Real-time Risk Assessment:** Future frameworks could implement continuous monitoring systems to update risk profiles as borrower data changes. This can help in proactively managing risk and adjusting lending policies accordingly.